Intelligent Edge-AI Medicine Dispenser

Hardware Platform: Arduino UNO Q (Qualcomm Dragonwing QRB2210 + STM32U585 MCU)

Development Environment: Arduino App Lab

1. Project Overview & Architectural Philosophy

The Intelligent Edge-AI Medicine Dispenser is an autonomous, fully offline medication management system. It transcends the capabilities of conventional pill organizers by functioning as an intelligent healthcare assistant that understands prescriptions, interacts naturally with patients, prevents overdoses, and mechanically extracts medications directly from their original uncut blister strips.

The system is built upon the dual-processor architecture of the **Arduino UNO Q**. This heterogeneous computing approach allows the system to separate high-level artificial intelligence (AI) and computer vision (CV) workloads from deterministic, hard real-time mechanical control:

- **The Linux MPU (Qualcomm Dragonwing):** Handles natural language processing, localized Large Language Models (LLMs), database management, and intensive computer vision pipelines.
- **The STM32 Microcontroller (MCU):** Dedicated exclusively to reliable, precise motor control, sensor monitoring, and safety limits, operating without interference from AI processing loads.

2. Advanced Functional Capabilities

A. Conversational Healthcare Assistant (Offline LLM)

Unlike cloud-dependent voice assistants, the dispenser operates entirely offline using a lightweight, quantized LLM (e.g., Llama.cpp deploying TinyLlama).

- **Natural Language Interface:** Patients interact conversationally rather than using fixed commands.
- **Query Resolution:** Patients can ask, "Did I take my morning pills?" and the LLM will query the local database to provide an accurate, timestamped answer.
- **Proactive, Contextual Check-ins:** The AI acts dynamically based on prerequisites. It may ask, "It is noon. Have you finished your lunch?" and postpone dispensing until the patient confirms they have eaten, if the medication requires it.
- **Pharmacological Guidance:** The local AI acts as a medical encyclopedia, answering queries about drug-to-drug interactions ("Can I take these two medicines together?") or explaining side effects.

B. Intelligent Scheduling, Inventory, & Overdose Prevention

- **Prescription OCR & Auto-Scheduling:** Using the onboard camera, Optical Character Recognition (OCR) extracts drug names, dosages, frequencies, and durations from printed prescriptions to automatically generate medication schedules in the local database.
- **Overdose Prevention:** Before any dispensing event, the AI checks the previous dispensing time, minimum dosing interval, and daily limits. It actively refuses to dispense medication if doing so would exceed safe thresholds, informing the user of the next permitted window.
- **Inventory Tracking & Alerts:** The system continuously monitors the number of remaining tablets. Every successful dispensing operation updates the inventory database, calculating the days of medication left and the next refill date.

C. Comprehensive Vision & Smart Display System

- **On-Device Smart Display:** A built-in mini screen (TFT/LCD) acts as the machine's primary visual interface. It provides at-a-glance information including the number of pills remaining, the exact time the last medication was taken, and renders simple visual graphs showing the patient's pill consumption frequency.
- **Face Recognition Authentication:** To prevent accidental misuse, medication is dispensed only after local facial recognition verifies the authorized patient or registered caregiver.
- **Medicine Identification:** The camera scans loose pills or blister packs to identify them by visual characteristics before they are loaded, preventing mix-ups.
- **Visual Cross-Check (Caretaker Mode):** The machine displays a caregiver's pre-captured reference photo of the required dose on its built-in screen during dispensing, allowing the patient to manually verify their medication.
- **Action Recognition:** Computer vision tracks the patient's hand-to-mouth movement, providing a stronger indication of actual medication consumption rather than merely logging that a pill was dispensed.
- **Camera-Driven Behavior Analysis:** The camera feed proactively monitors medication routines. It determines *why* pills are skipped (e.g., patient absent vs. patient refusal) and uses this context to initiate verbal check-ins regarding side effects and alerts the caretaker dashboard.

D. Vision-Guided Deblistering Mechanics (Core Innovation)

The system eliminates the need for manual pill sorting by extracting tablets directly from their original, uncut aluminum blister strips, ensuring pharmaceutical integrity.

Front-to-Back Coordinate Mapping: The camera observes the transparent front of the inserted blister strip, detects the exact geometric position of each tablet, and mathematically transforms those X/Y coordinates to the opaque aluminum foil side.

- **Adaptive U-Shaped Scoring Mechanism:** A 2-axis CNC-style stepper gantry aligns a precision cutter to create a 270-degree U-shaped score (a hinged flap) in the aluminum backing, avoiding the high-force punching that damages fragile tablets.
- **Dynamic Safety Margins:** The software automatically calculates a clearance buffer (e.g., pill radius + 2mm) to ensure the cutting tool never touches the medication.
- **Soft Plunger Dispensing:** A silicone plunger gently ejects the tablet through the flap.
- **Closed-Loop Dispensing Verification:** Post-dispense, the camera verifies the pocket is empty, the foil opened correctly, and the tablet successfully entered the delivery chute.

E. Telemetry & Cloud Integration (When Online)

While fully functional offline, the system offers secure, intermittent synchronization when Wi-Fi is available.

- **Caretaker Dashboard:** Syncs adherence logs, missed doses, and remaining inventory to a secure web/mobile dashboard.
- **Automated Refill Alerts:** Sends notifications to caregivers or local pharmacies when a specific medication is running low.

- **Behavioral Alerts:** Flags erratic behaviors detected by the camera (e.g., failed facial recognitions or repeated skipped doses) for caregiver review.

3. Open-Hardware & Community Ecosystem

The project is designed to be open and reproducible, acting as an educational and research platform for assistive healthcare.

- **Reproducibility:** 3D-printable mechanical components for the chassis, strip holder, and dispensing gantry, designed in free parametric CAD.
- **Modular Architecture:** Open PCB designs and modular Arduino/Python source code allow researchers to replace the AI model, camera, or sensors without redesigning the entire system.
- **Community Medicine Database:** An expandable database where the community can upload and share new blister pack geometries and dispensing parameters.

4. Technology Stack & Tooling

To execute this integrated system, the following technologies and skill sets must be utilized across the project lifecycle:

A. Development Environment & Systems Logic

- **Arduino App Lab:** The unified IDE for writing Python AI scripts, C++ motor sketches, and integrating edge AI "Bricks" across the dual-processor board.
- **Python:** The primary language for the overarching state machine logic.
- **Arduino RPC (Remote Procedure Call):** Manages secure Inter-Process Communication (IPC) between the Linux MPU and the STM32 MCU.
- **SQLite / TinyDB:** Lightweight, offline Python databases for maintaining schedules, user profiles, and medication history.

B. Edge AI & Computer Vision

- **llama.cpp & Whisper.cpp / Vosk:** Highly optimized C++ runtimes for deploying quantized LLMs and offline Speech-to-Text on the ARM Cortex-A CPU.
- **OpenCV & MediaPipe:** Used for Face Recognition, Pose/Action tracking, OCR, and the mathematical coordinate transformations required for the vision-guided cutting gantry.
- **Edge Impulse (YOLO-nano):** Integrated within App Lab to train and deploy lightweight object detection models for pill and blister pack identification.

C. Firmware & Hardware Control

- **Arduino C++ (STM32 MCU):** Programming the real-time execution loop.
- **AccelStepper Library:** Essential for non-blocking, smooth stepper motor control, allowing the MCU to drive the CNC gantry without freezing.
- **TFT/LCD UI Libraries (e.g., LVGL):** Renders the interactive dashboard, graphs, and cross-check images on the device's screen.

D. Mechanical CAD & 3D Printing

- **Parametric CAD (FreeCAD / Onshape / Fusion 360):** Used to design the complex 2-axis CNC gantry, the mechanical plunger, and the blister strip hoppers, utilizing variable-driven dimensions for precise tolerance adjustments.

- **Slicer Software:** UltiMaker Cura or PrusaSlicer to generate G-code for 3D printing the mechanical chassis.

E. Cloud Telemetry & Caregiver Dashboard

- **Backend:** Firebase or Supabase for secure, real-time database syncing and user authentication.
- **Frontend:** Next.js (React) for a responsive web dashboard or Flutter for mobile deployment.
- **Protocol:** MQTT or standard Python REST API requests to securely push structured JSON payloads when online.